b Four from:

- It is used to examine the 'traffic' between a user's computer (or internal network) and a public network (for example, the internet).
- It checks whether incoming or outgoing data meets a given set of criteria, and if the data fails the criteria, the firewall will block the 'traffic' and give the user (or network manager) a warning that there may be a security issue.
- The firewall can be used to log all incoming and outgoing 'traffic' to allow later interrogation by the user (or network manager).
- Criteria can be set so that the firewall prevents access to certain undesirable sites; the firewall can keep a list of all undesirable IP addresses.
- Alternatively, the firewall can be set to only allow access to known and trusted IP addresses.
- It is therefore possible for firewalls to help prevent viruses or hackers from entering the user's computer (or internal network).
- The user is warned if some software on their system is trying to access an external data source (for example, automatic software upgrade); the user is given the option of allowing it to go ahead or request that such access is denied.

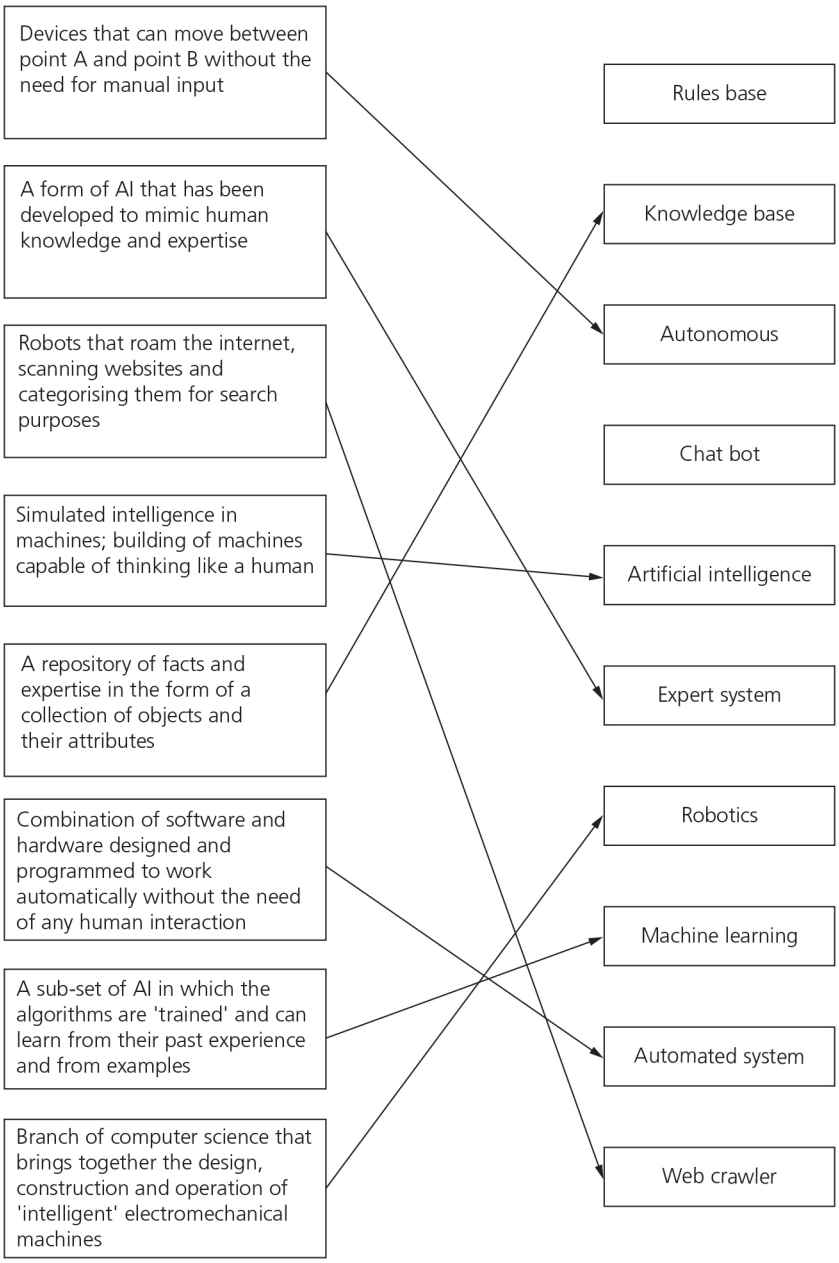
6 Automated and emerging technologies

1

- expert systems
- inference
- user interface
- dialogue boxes
- explanation system
- inference engine
- knowledge base
- inference engine
- inference rules
- rules base
- objects
- attributes
- repository

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2



3 **a** It is a combination of software and hardware designed and programmed to work automatically without the need for human intervention.

b An input device that takes readings from its surroundings and sends the data to a microprocessor/computer.

c Three from:

- industrial uses
- transport systems
- agriculture/farming
- weather forecasting
- gaming
- lighting systems
- science/research/laboratory work

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d Three from:

- They are much faster than a human operator to take any necessary action.
- They keep humans away from potentially dangerous environments.
- The process can be run under optimum conditions.
- They are less expensive in the long run.

e Three from:

- They are expensive to set up initially.
- It is always possible for a situation to arise which was not considered when testing and developing the system.
- They are subject to cyber attacks.
- They require enhanced maintenance.

4 a OCR

b

- A pressure sensor on the ground recognises presence of a vehicle.
- The data is sent back to computer which then sends a signal …
- … to operate the camera.
- The camera photographs the front of the vehicle.
- Software converts characters on the number plate into electronic values which are then compared to a database.
- If the vehicle number plate matches data on the database, then the computer sends a signal to an actuator …
- … to operate a motor to raise the barrier.
- On exit, the same procedure as above is carried out to see if the vehicle was allowed to park.

c i

- There is no need for any manual checks.
- It can operate 24/7.
- It has improved security because only recognised vehicles are allowed entry.

ii

- The number plate stored with manufacturer of vehicle/colour of vehicle is also stored on the database.
- There will be a vehicle manufacturer badge as well as number plate read by camera.
- It is possible to have additional security such as RFID in a vehicle.

5 a i cameras:

- can gauge the size of the parking space
- feed back images to driver to check parking procedure

ii sensors:

- can detect the presence of other vehicles
- can detect items such as kerbs, people, lamp posts, trees, and so on

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iii actuators:

- operate brakes
- operate steering
- operate throttle/accelerator

b To prevent the driver exceeding a set speed:

- Sensors in the front bumper of car detect the vehicle in front.
- Data is sent to the microprocessor (converted to digital by ADC if necessary).
- The microprocessor calculates the distance between vehicles.
- Sensors on the speedometer drive constantly send signals to the microprocessor.
- The microprocessor compares the speed to a set speed.
- If speed is equal to set speed no action is taken.
- If speed > set speed, signals are sent to actuators to operate brakes and reduce throttle.
- If speed < set speed, microprocessor sends signals to actuators to increase throttle.
- This process loop is continued until speed is within set speed.
- To prevent constant braking and accelerating, the microprocessor allows a 5 kph margin in all calculations.

To prevent the car getting too close to the vehicle in front of it:

- The microprocessor determines if the distance between vehicles is equal to the safe distance for set speed calculated above.
- If distance < safe distance, the microprocessor sends signals to operate brakes and reduce throttle …
- … and signal is also sent to a warning buzzer to inform the driver.
- If distance >= safe distance, the microprocessor checks speed value …
- … if speed = set speed then no action is taken …
- … otherwise, the first loop is checked to see if the car is doing the set speed value.
- This process loop is continued until vehicle is at a safe distance …
- … to prevent constant braking, a safety margin is built into calculations.

6 a i

- level sensor – to measure the amount of liquid added to a batch
- temperature sensor – to ensure the reaction mixture has the correct temperature of 35°C maintained
- light sensor – intensity of light passing through a reaction mixture will change as the colour changes

ii

- infrared/motion sensor – detects the presence of visitors in the room
- light sensor – to automatically switch off room lights once the display starts

iii

- pressure sensor – to detect a passenger in a doorway
- infrared sensor – to detect the proximity of a passenger
- camera – to relay data back to the driver or to the control centre

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b

Statements	True (✓)	False (✓)
Automated systems lead to less consistent results or less consistent products		✓
Automated systems are more expensive to set up than traditional manual systems	✓	
Automated systems could be quickly overwhelmed by the amount of data presented to it		✓
Automated systems are inherently less safe than manual systems		✓
Automated systems generally require enhanced maintenance when compared to manual systems	✓	
Automated systems allow processes to run at optimum conditions at all times	✓	
Software failures, due to unforeseen conditions, are unlikely to impact on an automated system		✓
Automated systems will react more quickly to unusual process conditions than a manual system	✓	

7 a 1 = user interface

2 = inference engine

3 = knowledge base

b i objects

ii taxi cab

c

- Information needs to be gathered from human experts or from written sources such as textbooks, research papers or the internet.
- Information gathered is used to populate the knowledge base which first needs to be created.
- A rules base needs to be created; this is made up of a series of inference rules so that the inference engine can draw conclusions.
- The inference engine itself needs to be set up; it is a complex system because it is the main processing element making reasoned conclusions from data in the knowledge base.
- The user interface needs to be developed to allow the user and expert system to communicate.

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8 a i the simulation of human behaviour – such as recognising objects, problem-solving, decision-making and language skills – by a computer, essentially:

- a collection of rules and data
- the ability to reason, learn and adapt to external stimuli

ii a sub-set of AI, in which algorithms are trained to learn from past experiences and examples

b

AI	**Machine learning**
It represents simulated intelligence in machines	This is the practice of getting machines to make decisions without being programmed to do so
The aim is to build machines which are capable of thinking like humans	The aim is to make machines that learn through data acquisition, so that they can solve new problems

9 a C

b E

c B

d A

e D

10

Advantages of autonomous trains	**Disadvantages of autonomous trains**
improved punctuality	the ever-present fear of hacking into the vehicle's control system
reduced running costs (fewer staff are required)	system doesn't work well with very busy services (at the moment)
improved safety because human error is removed	high capital costs and operational costs initially (that is, buying the trains, expensive signalling and control equipment and the need to train staff)
minimises energy consumption because there is better control of speed and provides for a more optimum service with fewer delays; delayed trains stuck in stations still use energy	ensuring that passenger behaviour is acceptable particularly during busy times (for example, jamming doors open on trains, standing too near the edge of platforms, and so on)
it is possible to increase the frequency of trains (automated systems allow for shorter times between trains)	passenger reluctance of the new technology
it is easier to change train scheduling (for example, more trains during busier times)	no drivers mean there will be a need for CCTV to monitor railway stations

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Advantages of autonomous cars and buses	Disadvantages of autonomous cars and buses
safer because human error is removed leading to fewer accidents	very expensive system to set up in the first place (high technology requirements)
better for the environment because vehicles will operate more efficiently	the ever-present fear of hacking into the vehicle's control system
reduced traffic congestion (humans cause 'stop-and-go' traffic known as 'the phantom traffic jam' whereas autonomous vehicles will be better at smoothing out traffic flow, reducing congestion in cities)	security and safety issues (software glitches could be catastrophic; software updates would need to be carefully controlled to avoid potential disasters)
increased lane capacity (research shows that autonomous vehicles will increase lane capacity by 100% and increase average speeds by 20%, due to better braking and acceleration responses together with optimised distance between vehicles)	the need to make sure the system is well-maintained at all times; cameras need to be kept clean so that they don't give false results; sensors could fail to function in heavy snowfall or blizzard conditions (radar or ultrasonic signals could be deflected by heavy snow particles)
reduced travel times (for the reasons above) therefore less commuting time	driver and passenger reluctance of the new technology
stress-free parking for motorists (the car will find car parking on its own and then self-park)	reduction in the need for taxis could lead to unemployment (imagine New York without its famous yellow cabs!)

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